

Christopher Turner

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Summary

- Accomplished materials research and development for next-generation applications.
- Thorough knowledge of advanced laboratory techniques; solid state, high temp., high and low pressure, characterization, thin film, sintering, mechanical alloying.
- Experience with high-throughput production machining environments from materials received to finished product.
- Exceptional multi-tasking capabilities. Possess working knowledge of project management, systems diagnostics and organization, both technical and administrative; material sourcing/securing, process development, and production.

Education

University of California, Los Angeles
Ph.D., Inorganic Chemistry

Los Angeles, CA
June 2016

University of California, Los Angeles
M.Sc., Materials Chemistry

Los Angeles, CA
March 2014

University of California, Los Angeles
B.Sc., Chemistry

Los Angeles, CA
June 2012

Experience

Supermetalix, Inc.
Chief Science Officer

Chico, CA
Dec 2016 - Present

- Directing research on development of IP / patented materials for industrial use
- Investigation of scalability and processing of feed stock from raw materials
- Research of industry competitive sintered ceramic-metallic composites for tooling
- Sourcing and evaluating feed stock materials for manufacturing of products
- Design and fabrication of devices and instrumentation for scaling and production
- Collaboration with industry experts on commercial applications

JPL Visiting Scholar Research Program
Visiting Researcher

Pasadena, CA
Jun 2015 – Dec 2016

- Investigation of novel ceramics for thermal and diffusion barriers
- Process and efficiency optimization for space application thermo-electric materials
- Developed different approaches for materials synthesis scalability / consistency
- Design and fabrication of reaction vessels and instrumentation

UCLA, Dept. of Chemistry and Biochemistry
Graduate Researcher

Los Angeles, CA
Sept 2012 – Dec 2016

- Investigation of metal-matrix composites—Synthesis and Application
- Solid State Chemistry research investigating Superhard, Refractory Transition-Metal Borides
- High temperature / high pressure synthesis of new materials
- Advanced characterization techniques applied to metal borides for structure and electronic properties

Undergraduate Researcher

- Formulation and preparation of raw materials for synthesis
- High throughput boride phase screening under the direction of senior researchers
- Maintenance and retrofitting of laboratory equipment and testing apparatus

Consulting

California

9/2008 - Present

- Design and modeling components used in aerospace, automotive, food, and science industries
- Materials Selection and Processing
- Manufacturing optimization for new and existing production environments
- Process engineering and development of chemical systems

Skillset

Special Technical Training

(Materials Selection / Characterization and Processing)

- Atomic Force Microscopy
- Energy Dispersive X-ray Spectroscopy
- Scanning Electron Microscopy
- S.Q.U.I.D. Magnetometry
- X-Ray Diffraction
- TGA/DTA
- UV/VIs/IR/Near-IR Spectroscopy
- Powder Metallurgical Processes
- High Vac. Systems
- High Temperature Systems
- Spark Plasma Sintering / Hot Pressing

Drafting/Design/CAD/CAM

- Solidworks Platform
- Autodesk Systems - AutoCAD
- Gibbscam CAD/CAM software

Special Non-Technical Training

- Laboratory Safety
- Radiation Safety Training
- Laser Safety Training
- Hazmat Safety Training
- Machine Safety Training
- Safety Protocol Development

Publications

Patents / Applications

Metal Borides and uses thereof

US Patent App. 16/183,346

Mixed Metal Dodecaborides and uses thereof

US Patent 11,168,001

Tungsten tetraboride composite matrix and uses thereof

US Patent 11,174,538

Refereed Journals

Enhanced Hardening Effects on Molybdenum-Doped WB₂ and WB₂-SiC/B₄C Composites

L.E. Pangilinan, S. Hu, C.L. Turner, J. Yan, A. Kavner, R. Mohammadi, S.H. Tolbert, R.B. Kaner

Chem. Mater. 34 (12), 5461-5470, 2022

Synthesis and high-pressure mechanical properties of superhard rhenium/tungsten diboride nanocrystals

J. Lei, S. Hu, C.L. Turner, K. Zeng, M.T. Yeung, J. Yan, R.B. Kaner, S.H. Tolbert

ACS nano 13 (9), 10036-10048, 2019

Fast response electrochemical capacitor electrodes created by laser-reduction of carbon nanodots

V. Strauss, M. Anderson, C.L. Turner, R.B. Kaner

Materials Energy Today 11 114-119, 2019

Microscopic investigation of local structural and electronic properties of tungsten tetraboride: a superhard metallic material

C.L. Turner, Z. Zukovic, D. Koumoulis, R.E. Taylor, R.B. Kaner

Journal of Materials Science 54 (4), 3547-3557 2019

Synthesis and High Pressure Mechanical Properties of Superhard Rhenium/Tungsten Diborides Nanocrystals

J. Lei, S. Hu, C.L. Turner, K. Zeng, M.T. Yeung, J. Yan, R.B. Kaner, S.H. Tolbert

ACS Nano, doi.org/10.1021/acsnano.9b02103, 2019

Fast response electrochemical capacitor electrodes created by laser-reduction of carbon nanodots

V. Strauss, M. Anderson, C.L. Turner, R.B. Kaner

Materials Today Energy 11, 114-119 2019

Synthesis and characterization of aluminum diboride products using ^{27}Al , ^{11}B NMR and ab initio studies

C.L. Turner, D. Koumoulis, G. Li, Z. Zujovic, R.E. Taylor, R.B. Kaner

Journal of Materials Science 53 (5), 3309-3322 2018

Understanding How Bonding Controls Strength Anisotropy in Hard Materials by Comparing the High-Pressure Behavior of Orthorhombic and Tetragonal Tungsten Monoboride

J. Lei, M.T. Yeung, P.J. Robinson, R. Mohammadi, C.L. Turner, J. Yan, R.B. Kaner

The Journal of Physical Chemistry C 122 (10), 5647-5656, 2018

Effects of Variable Boron Concentration on the Properties of Superhard Tungsten Tetraboride

G. Akopov, I. Roh, Z.C. Sobell, M.T. Yeung, L. Pangilinan, C.L. Turner, R.B. Kaner

Journal of the American Chemical Society 139 (47), 17120-17127, 2017

^{11}B NMR Study of WB_2

C.L. Turner, Z. Zujovic, D. Koumoulis, R.E. Taylor, R.B. Kaner

The Journal of Physical Chemistry C 121 (2), 1315-1320, 2017

Superhard Rhenium/Tungsten Diboride Solid Solutions

A.T. Lech*, C.L. Turner*, J. Lei, R. Mohammadi, S.H. Tolbert, R.B. Kaner

Journal of the American Chemical Society 138 (43), 14398-14408, 2016

Superhard Mixed Transition Metal Dodecaborides

G. Akopov, M.T. Yeung, Z.C. Sobell, C.L. Turner, C.W. Lin, R.B. Kaner

Chemistry of Materials 28 (18), 6605-6612, 2016

Superhard Monoborides: Hardness Enhancement through Alloying in $\text{W}_{1-x}\text{Ta}_x\text{B}$

M.T. Yeung, J. Lei, R. Mohammadi, C.L. Turner, Y. Wang, S.H. Tolbert, R.B. Kaner

Advanced Materials 28 (32), 6993-6998, 2016

Stabilization of HfB_{12} in $\text{Y}_{1-x}\text{Hf}_x\text{B}_{12}$ under Ambient Pressure

G. Akopov, M.T. Yeung, C.L. Turner, R.L. Li, R.B. Kaner

Inorganic chemistry 55 (10), 5051-5055, 2016

Extrinsic hardening of superhard tungsten tetraboride alloys with group 4 transition metals

G. Akopov, M.T. Yeung, C.L. Turner, R. Mohammadi, R.B. Kaner

Journal of the American Chemical Society 138 (17), 5714-5721, 2016

^{11}B NMR Spectral and Nuclear Spin-Lattice Relaxation Analyses of ReB_2

D. Koumoulis, C.L. Turner, R.E. Taylor, R.B. Kaner

The Journal of Physical Chemistry C 120 (5), 2901-2907, 2016

Enhancing the hardness of superhard transition-metal borides: molybdenum-doped tungsten tetraboride

R. Mohammadi, C.L. Turner, M. Xie, M.T. Yeung, A.T. Lech, S.H. Tolbert, R.B. Kaner

Chemistry of Materials 28 (2), 632-637, 2016

Mesoporous Ni₆₀Fe₃₀Mn₁₀-alloy based metal/metal oxide composite thick films as highly active and robust oxygen evolution catalysts

E. Detsi, J.B. Cook, B.K. Lesel, C.L. Turner, Y.L. Liang, S. Robbennolt, S.H. Tolbert
Energy & Environmental Science 9 (2), 540-549, 2016

Fundamentals and Applications of the World's Hardest Metals

C.L. Turner

Doctoral Thesis – UCLA, 2016

Exploring hardness enhancement in superhard tungsten tetraboride-based solid solutions using radial X-ray diffraction

M. Xie, R. Mohammadi, C.L. Turner, R.B. Kaner, A. Kavner, S.H. Tolbert
Applied Physics Letters 107 (4), 041903, 2015

¹⁰B and ¹¹B NMR Study of Elemental Boron

C.L. Turner, R.E. Taylor, R.B. Kaner

The Journal of Physical Chemistry C 119 (24), 13807-13813, 2015

Structure of superhard tungsten tetraboride: A missing link between MB₂ and MB₁₂ higher borides

A.T. Lech, C.L. Turner, R. Mohammadi, S.H. Tolbert, R.B. Kaner
Proceedings of the National Academy of Sciences, 201415018, 2015

Lattice stress states of superhard tungsten tetraboride from radial x-ray diffraction under non-hydrostatic compression

M. Xie, R. Mohammadi, C.L. Turner, R.B. Kaner, A. Kavner, S.H. Tolbert
Physical Review B 90 (10), 104104, 2014

Vapor-phase polymerization of nanofibrillar poly (3, 4-ethylenedioxythiophene) for supercapacitors

J. D'Arcy, M. El-Kady, P. Khine, L. Zhang, S. Lee, N. Davis, D. Liu, M. Yeung, S. Kim, C.L. Turner, A. Lech, P. Hammond and R.B. Kaner
ACS Nano 8 (2), 1500-1510, 2014

Toward Inexpensive Superhard Materials: Tungsten Tetraboride-Based Solid Solutions

R. Mohammadi, M. Xie, A.T. Lech, C.L. Turner, A. Kavner, S.H. Tolbert, R.B. Kaner
Journal of the American Chemical Society, 134, 20660-20668, 2012